

20251010 First Glance at PHANGS-MUSE NGC5068

PHAGNS-MUSE nebula catalog: <https://www.eso.org/sci/publications/announcements/sciann17705.html>

Description: <https://www.eso.org/rm/api/v1/public/releaseDescriptions/235>

I checked the PHANGS-MUSE's released catalog, and it turns out that they have some emission lines and BPT classification. So then I pick the low mass and face-on galaxy NGC5068 ($\log(M_*/M_\odot) = 9.40$, $\text{deg} = 35.7^\circ$) to have a look:

1. They dont provided information about the stellar mass surface density so we can not do rMZR or rSFMS.
2. They provided corrected $H\alpha$ so we can may consider it as the Σ_{SFR} . Afterall they are just offset by a coefficient inside single galaxy.
3. They provided metallicity but it is calculated by Scal-PG16, so I recalculated the $[\text{O}/\text{H}]$ by O3N2-M13.
4. They also provided the BPT classification so I can make sure we only look at the HII region.

And in short, I think the results here rule out the possibility that the inclination may cause negative correlation between metallicity and Σ_{SFR} in low mass galaxies.

As usual, below is the $H\alpha$ and O3N2-M13 map, as well as the $Z_{\text{gas}}\text{-}H\alpha$ relation. Again, if you consider that $H\alpha$ represents the Σ_{SFR} , then we can clearly see the trend that higher Σ_{SFR} (redder on LHS), lower metallicity (bluer on RHS). And even though we dont have mass information to bin the $Z_{\text{gas}}\text{-}H\alpha$ relation, we can still clearly see the negative correlation.

